

COMPUTATIONAL EPIDEMIOLOGY

Agent-Based Pandemic Simulation

*A High-Fidelity SEIR-V Model with Spatial Dynamics,
Mutation Mechanics & Healthcare Constraints*

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PRESENTATION OVERVIEW

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- *Problem context & motivation*

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- *Findings, limits & future work*

INTRODUCTION

Traditional mathematical models often fail to capture the stochastic and spatial nature of real-world disease transmission.



The Problem

- *Pandemic outbreaks demand rapid, accurate modeling. Classical ODE-based SIR/SEIR models assume unrealistic homogeneous mixing.*



Our Approach

- *An Agent-Based Model (ABM) where each individual is autonomous, moves spatially, and interacts locally — capturing emergent epidemic behavior.*



The Goal

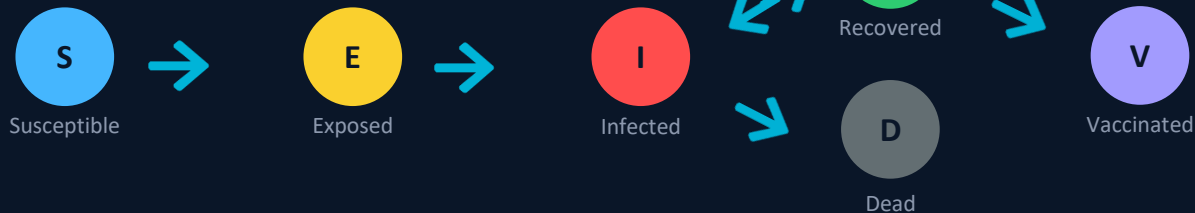
- *A flexible experimental platform for testing intervention strategies: vaccination timing, infection control, and variant emergence.*

THEORETICAL BACKGROUND

- **Classical Models**

- ! *Differential equation systems (SIR / SEIR)*
- ! *Assume homogeneous population mixing*
- ! *Mathematically elegant but spatially blind*
- ! *Cannot capture individual behavior or clustering*
- ! *No stochastic transmission events*

- **SEIR-V State Machine**



- **Agent-Based Modeling**

- ✓ *Each person is an autonomous agent*
- ✓ *Spatially constrained local interactions*
- ✓ *Stochastic infection & recovery events*
- ✓ *Heterogeneous demographics (age, chronic illness)*
- ✓ *Emergent outbreak clusters & wave dynamics*

SYSTEM ARCHITECTURE



Initialization

- *Random agent placement in 2D space.*
- *Age, chronic status & initial states assigned.*



Movement Engine

- *Trigonometric motion ($\cos/\sin$).*
- *Boundary reflection. Essential workers unrestricted.*



Transmission

- *cdist() pairwise distance matrix.*
- *Exposure if $d_{ij} < \text{radius}$ AND $p < P_{\text{inf}}$.*



State Transitions

- *Time-in-state counters trigger $S \rightarrow E \rightarrow I \rightarrow R/D$.*
- *Death risk shaped by chronic illness & overload.*



Vaccination

- *Daily vaccination of N susceptible agents*
- *starting from configurable day.*



Visualization

- *Real-time Matplotlib rendering.*
- *Stack plot, spatial map & recovery trends.*

SIMULATION PARAMETERS



Default Parameter Set

Parameter	Default	Effect
Infection Radius	2.0 units	Proximity threshold
Infection Probability	50%	Base transmission rate
Exposed Duration	5 days	Incubation period
Infected Duration	20 days	Illness duration
Death Probability	10%	Base mortality
Hospital Capacity	200	Overload threshold
Vaccination Rate	10/day	Daily immunizations
Variant Inf. Prob.	80%	Increased transmissibility

EPIDEMIOLOGICAL DYNAMICS

Infection Condition: $d_{ij} < r$ AND $\text{rand}() < P_{inf}$



Cluster Formation

- Infections propagate through localized spatial proximity, forming geographic clusters and wave-like spread — impossible to replicate with ODE models.



Hospital Overload

- When active cases exceed `HOSPITAL_CAPACITY`, death risk doubles dynamically. Nonlinear mortality creates healthcare system tipping points.



Variant Emergence

- A new variant activates on a configurable day. Infection probability jumps to 80%, duration extends to 35 days, triggering a second epidemic wave.



Vaccination Strategy

- Daily vaccination of susceptible agents begins on a user-defined day. Shrinks the *S* population and reduces the effective reproduction number R_e .



Infection Control

- On the control start day, agent movement speed drops to 20% of normal. Essential workers remain unrestricted, maintaining critical mobility.



Mortality Model

- Base death risk $\times 3.0$ for chronic conditions. $\times 2.0$ for hospital overload. Multiplicative risk model captures real-world vulnerability stratification.

ALGORITHMIC COMPLEXITY & PERFORMANCE

$$O(N_s \times N_i)$$

Distance Computation Complexity

*Worst case approaches $O(N^2)$
when most agents are susceptible or infected*

`scipy.spatial.distance.cdist()`



Compiled C implementation — no Python loops



Full pairwise distance matrix in one call



Scales to thousands of agents efficiently

- **Optimization Pathways**

Spatial Partitioning

Grid-based indexing reduces candidate pairs from $O(N^2)$ to $O(k)$ where k = local neighborhood size.

KD-Tree Search

Nearest-neighbor queries in $O(\log N)$. Ideal for sparse infection scenarios.

Parallel Computation

Distance matrix operations are embarrassingly parallel — GPU acceleration feasible with CuPy.

Vectorized Transitions

State updates use NumPy boolean masks across all agents simultaneously — eliminates inner Python loops.

EXPERIMENTAL SCENARIOS

01 Baseline



- *No interventions applied. Uncontrolled spread leads to rapid infection peak, high cumulative deaths, and healthcare system overload.*

Highest mortality

02 Early Vaccination



- *Vaccination begins before the infection peak. Susceptible pool shrinks rapidly, reducing R_e below 1 and flattening the curve significantly.*

Lowest peak infection

03 Late Intervention



- *Control measures introduced after hospital capacity is breached. Mortality multiplier already active. Deaths significantly higher than early intervention.*

Delayed but costly

04 Variant Emergence



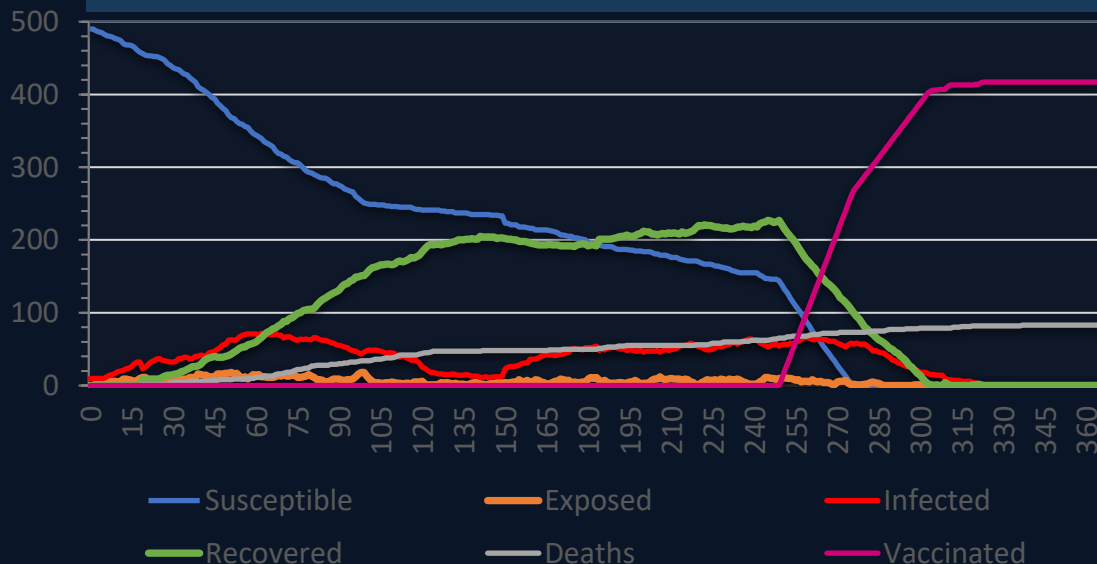
- *A new variant with 80% infection probability activates on day 150. Creates a distinct second epidemic wave with extended duration (35-day illness).*

Double-wave dynamics

All scenarios use identical agent populations — only timing and intervention parameters differ, isolating the causal effect of each strategy.

RESULTS & VISUALIZATION

SEIR-V Population Dynamics



Spatial Agent Map

Real-time scatter plot shows agent positions colored by state. Agent size encodes age. Chronic cases outlined in white. Infection clusters visible as red hotspots.



Stack Plot

Stacked area chart tracks all 6 population states (S/E/I/R/D/V) over time. Shows relative shifts and the impact of variant emergence as a second peak.



Recovery Trends

Separate lines for healthy vs. chronic recoveries reveal differential outcomes. Chronic patients recover at lower rates due to elevated death risk.



Snapshots & Report

Auto-saved PNG snapshots at key days (1, 50, 100, 150, 200, 250, 300, 364). Full time-series exported to `Pandemic_Report.xlsx` for post-hoc analysis.

LIMITATIONS & FUTURE ENHANCEMENTS

Current Limitations

Random Mobility

- *Agent movement is stochastic trigonometric — not calibrated to real human mobility patterns (commute, social gathering).*

No Social Networks

- *Interactions are purely proximity-based. Real populations have structured contact networks (households, workplaces, schools).*

Fixed Parameters

- *Parameters remain constant within a run. Real epidemics feature evolving transmission rates, seasonality, and behavioral change.*

Single Region

- *The simulation operates in one bounded 2D space. Cross-region spread and travel dynamics are not modeled.*

Future Enhancements



Real Epidemiological Data

- *Calibrate model parameters from WHO / ECDC datasets. Validate against COVID-19 or influenza outbreak records.*



AI-Based Prediction

- *Train an LSTM or transformer on simulation outputs to predict epidemic trajectories from early signals.*



Multi-Region Simulation

- *Implement inter-region agent mobility. Model cross-border travel and city-level epidemic coupling.*



Behavioral Modeling

- *Integrate risk-perception-driven behavior changes: mask adoption, voluntary isolation, social distancing compliance.*

CONCLUSION

A Powerful Foundation for Computational Epidemiology

- 1 *Agent-based modeling captures epidemic dynamics that ODE models fundamentally cannot — spatial clusters, individual heterogeneity, and stochastic transmission.*
- 2 *Intervention timing is the single most consequential variable: early vaccination and control measures dramatically reduce both peak infections and total mortality.*
- 3 *Healthcare capacity is a nonlinear threshold — exceeding it triggers a multiplicative mortality effect, underscoring the critical importance of 'flattening the curve'.*
- 4 *The SEIR-V framework is extensible: variant dynamics, demographic risk stratification, and vaccination rollout are already embedded in the architecture.*
- 5 *The simulation serves as both an educational tool and a rigorous research platform, with full data export for external validation and analysis.*



Thank you